```
#Name: Mohd. Affan Ansari ; UIN: 231A023
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
import pandas as pd
```

```
#Name: Mohd. Affan Ansari ; UIN: 231A023
documents = [
    "Artificial intelligence allows machines to solve complex problems.",
    "Machines can solve complex problems using artificial intelligence.",
    "Computers can process and understand human language.",
    "The students visited the museum during their college trip.",
    "The weather was pleasant throughout the morning."
]
```

```
#Name: Mohd. Affan Ansari ; UIN: 231A023
for i, document in enumerate (documents):
 print (f" D{i + 1}: {document}")
```

```
 D1: Artificial intelligence allows machines to solve complex problems.
 D2: Machines can solve complex problems using artificial intelligence.
 D3: Computers can process and understand human language.
 D4: The students visited the museum during their college trip.
 D5: The weather was pleasant throughout the morning.
```

```
#Name: Mohd. Affan Ansari ; UIN: 231A023
vectorizer = TfidfVectorizer(stop_words='english')
tfidf_matrix = vectorizer.fit_transform(documents)
print(tfidf_matrix.shape)
```

```
(5, 21)
```

```
#Name: Mohd. Affan Ansari ; UIN: 231A023
tfidf_df = pd.DataFrame(
tfidf_matrix.toarray(),
columns=vectorizer.get_feature_names_out(),
index=["D1", "D2", "D3", "D4", "D5"]
)
tfidf_df.round(2)
```

	allows	artificial	college	complex	computers	human	intelligence	language	machines	morning	...	pleasant	problems	p
D1	0.45	0.36	0.00	0.36	0.00	0.00	0.36	0.00	0.36	0.00	...	0.00	0.36	
D2	0.00	0.36	0.00	0.36	0.00	0.00	0.36	0.00	0.36	0.00	...	0.00	0.36	
D3	0.00	0.00	0.00	0.00	0.45	0.45	0.00	0.45	0.00	0.00	...	0.00	0.00	
D4	0.00	0.00	0.45	0.00	0.00	0.00	0.00	0.00	0.00	0.00	...	0.00	0.00	
D5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.58	...	0.58	0.00	

5 rows × 21 columns

```
#Name: Mohd. Affan Ansari ; UIN: 231A023
print(vectorizer.get_feature_names_out())
```

```
['allows' 'artificial' 'college' 'complex' 'computers' 'human'
 'intelligence' 'language' 'machines' 'morning' 'museum' 'pleasant'
 'problems' 'process' 'solve' 'students' 'trip' 'understand' 'using'
 'visited' 'weather']
```

```
#Name: Mohd. Affan Ansari ; UIN: 231A023
similarity_matrix = cosine_similarity(tfidf_matrix)

similarity_matrix
```

```
array([[1.        , 0.79614704, 0.        , 0.        , 0.        ],
       [0.79614704, 1.        , 0.        , 0.        , 0.        ],
       [0.        , 0.        , 1.        , 0.        , 0.        ],
       [0.        , 0.        , 0.        , 1.        , 0.        ],
       [0.        , 0.        , 0.        , 0.        , 1.        ]])
```

```
similarity_matrix = cosine_similarity(tfidf_matrix)
similarity_matrix
```

```
array([[1.        , 0.79614704, 0.        , 0.        , 0.        ],
       [0.79614704, 1.        , 0.        , 0.        , 0.        ],
       [0.        , 0.        , 1.        , 0.        , 0.        ],
       [0.        , 0.        , 0.        , 1.        , 0.        ],
       [0.        , 0.        , 0.        , 0.        , 1.        ]])
```

```
#Name: Mohd. Affan Ansari ; UIN: 231A023
similarity_df = pd.DataFrame(
similarity_matrix,
index=["D1", "D2", "D3", "D4", "D5"],
columns=["D1", "D2", "D3", "D4", "D5"]
)
similarity_df.round(3)
```

	D1	D2	D3	D4	D5
D1	1.000	0.796	0.0	0.0	0.0
D2	0.796	1.000	0.0	0.0	0.0
D3	0.000	0.000	1.0	0.0	0.0
D4	0.000	0.000	0.0	1.0	0.0
D5	0.000	0.000	0.0	0.0	1.0

```
#Name: Mohd. Affan Ansari ; UIN: 231A023
highest_similarity = -1
most_similar_pair = None
for i in range(len(documents)):
    for j in range(i + 1, len(documents)):
        similarity = similarity_matrix[i][j]
        if similarity > highest_similarity:
            highest_similarity = similarity
            most_similar_pair = (i, j)
print(
    f"Most similar documents: D{most_similar_pair[0]+1} and D{most_similar_pair[1]+1}"
)
print(f"Similarity: {highest_similarity:.3f}")
```

```
Most similar documents: D1 and D2
Similarity: 0.796
```